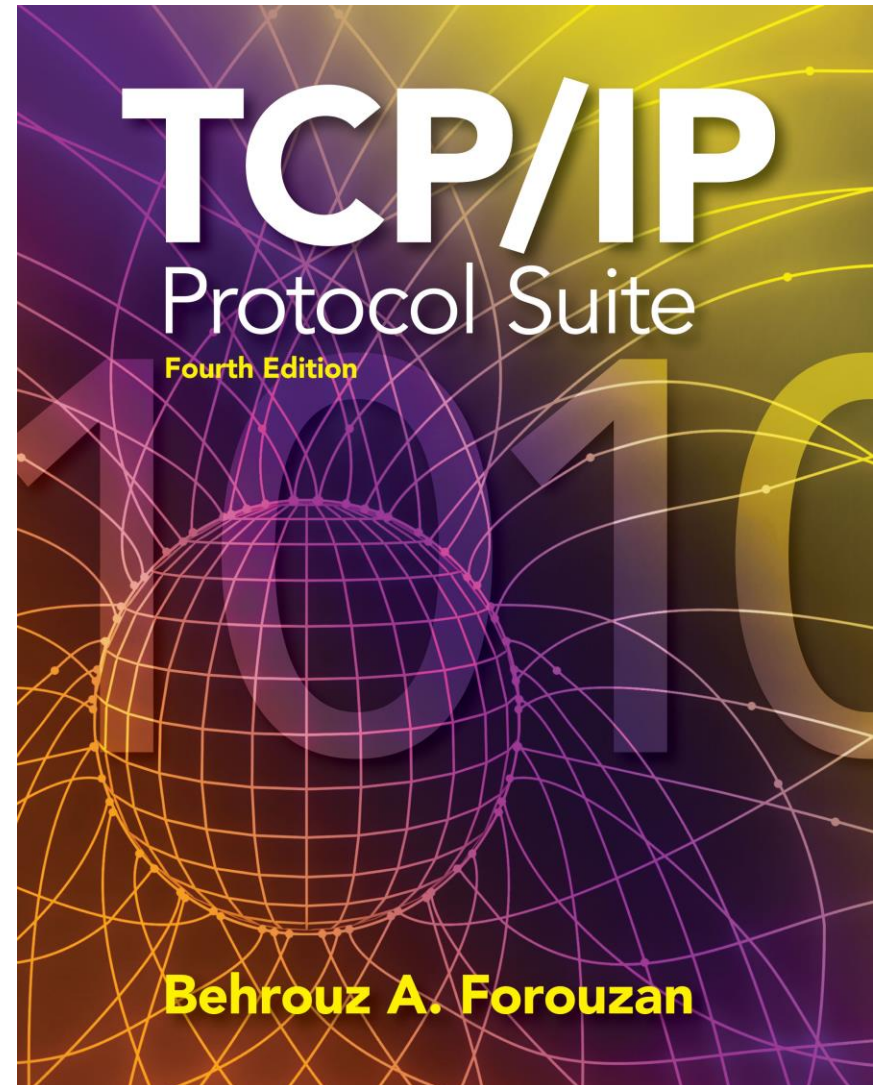


Chapter 10

Mobile IP



OBJECTIVES:

- ☐ To discuss addressing issues related to a mobile host and the need for a care-of address.
- ☐ To discuss two agents involved in mobile IP communication, the home agent and the foreign agent, and how they communicate.
- ☐ To explain three phases of communication between a mobile host and a remote host: agent discovery, registration, and data transfer.
- ☐ To mention inefficiency of mobile IP in two cases, double crossing and triangular routing, and a possible solution.

Chapter Outline

10.1 Addressing

10.2 Agents

10.3 Three Phases

10.4 Inefficiency in Mobile IP

10-1 ADDRESSING

The main problem that must be solved in providing mobile communication using the IP protocol is addressing.

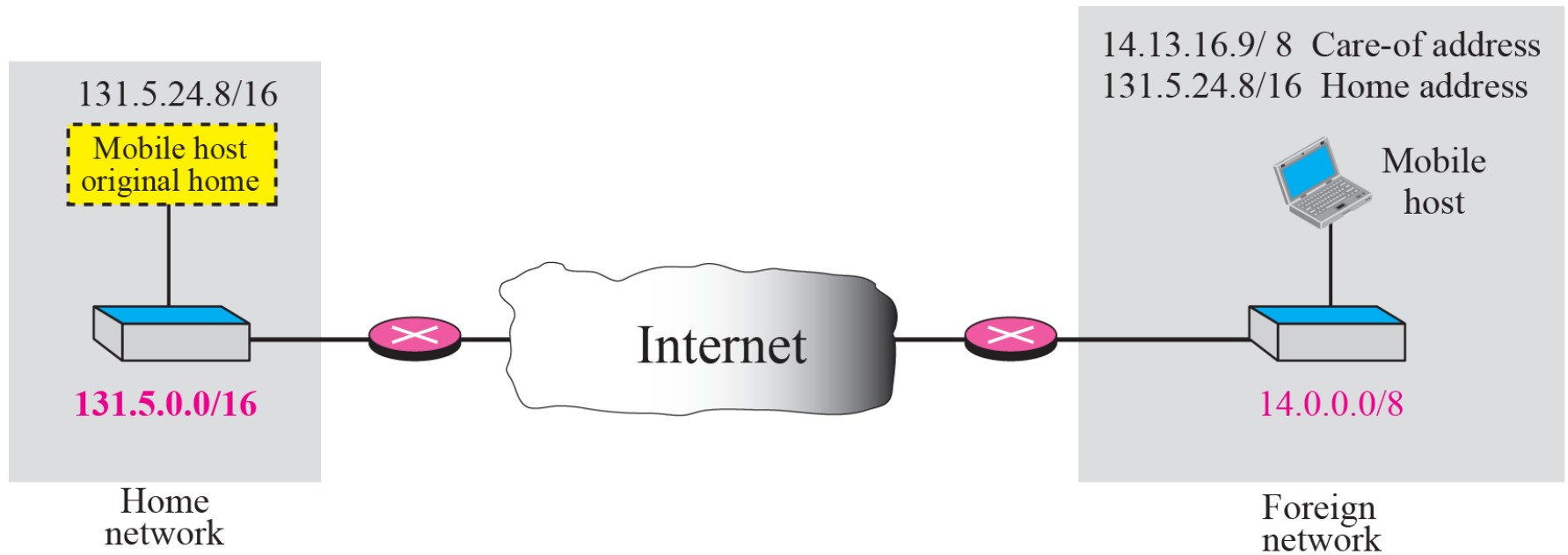
Topics Discussed in the Section

- ✓ **Stationary Host**
- ✓ **Mobile Host**

Note

The IP addresses are designed to work with stationary hosts because part of the address defines the network to which the host is attached.

Figure 10.1 *Home address and care-of address*



Note

Mobile IP has two addresses for a mobile host: one home address and one care-of address.

The home address is permanent; the care-of address changes as the mobile host moves from one network to another.

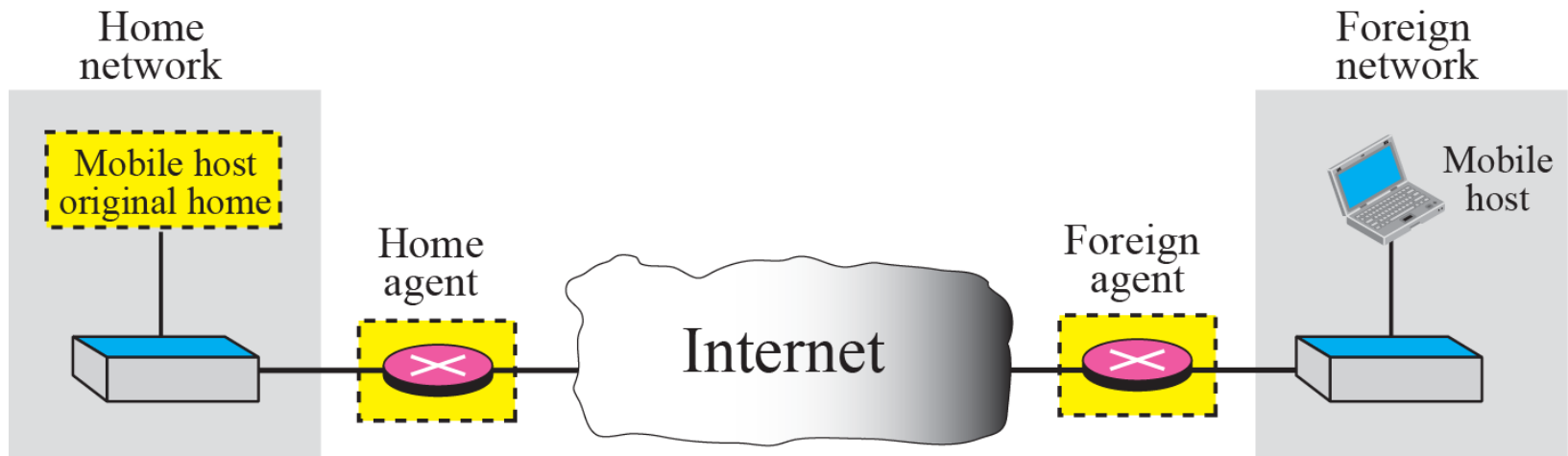
10-2 AGENTS

To make the change of address transparent to the rest of the Internet requires a home agent and a foreign agent.

Topics Discussed in the Section

- ✓ **Home Agent**
- ✓ **Foreign Agent**

Figure 10.2 *Home agent and foreign agent*



Note

When the mobile host and the foreign agent are the same, the care-of address is called a colocated care-of address.

10-3 THREE PHASES

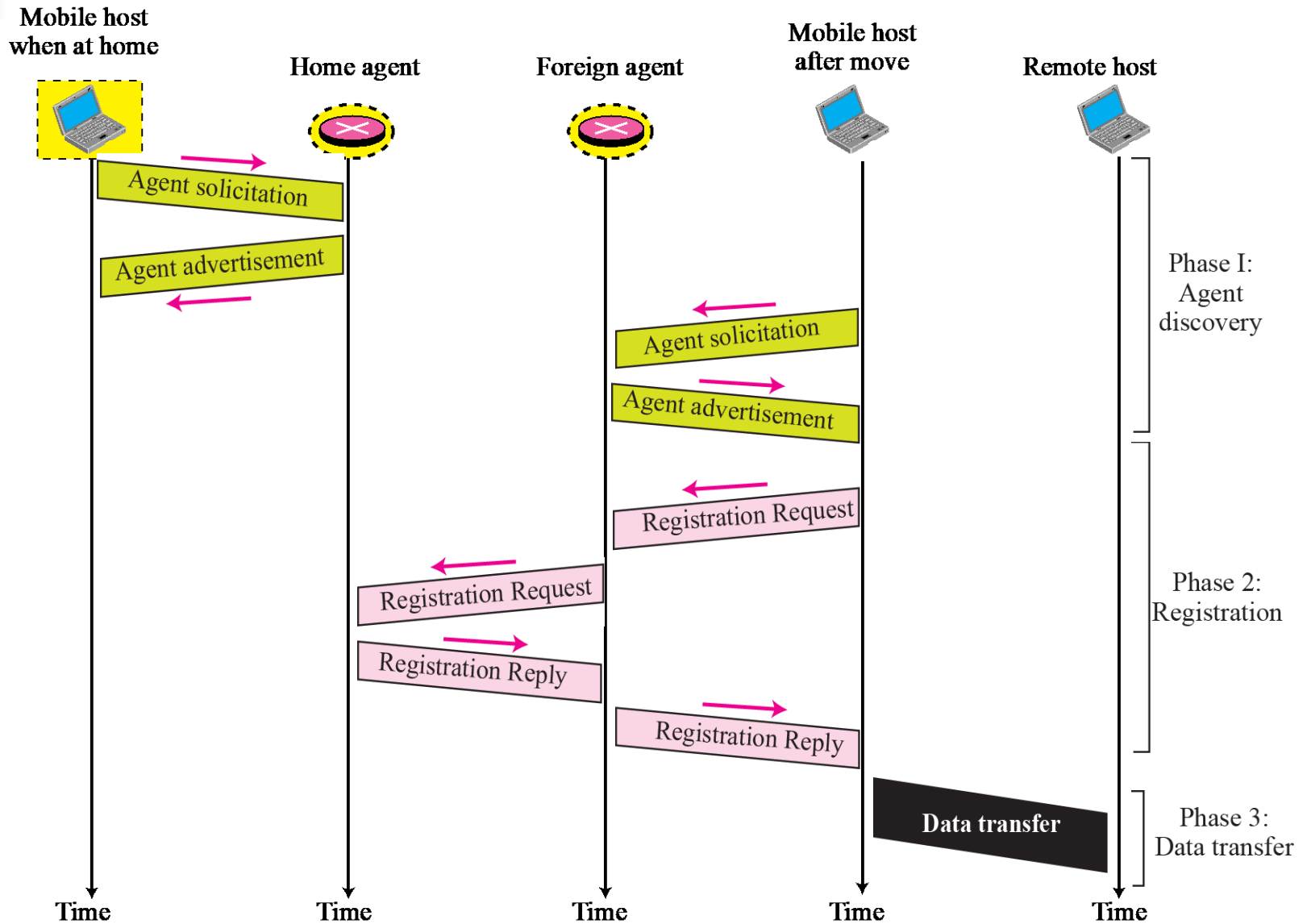
To communicate with a remote host, a mobile host goes through three phases: agent discovery, registration, and data transfer.

The first phase, agent discovery, involves the mobile host, the foreign agent, and the home agent. The second phase, registration, also involves the mobile host and the two agents. Finally, in the third phase, the remote host is also involved. We discuss each phase separately.

Topics Discussed in the Section

- ✓ **Agent Discovery**
- ✓ **Registration**
- ✓ **Data Transfer**

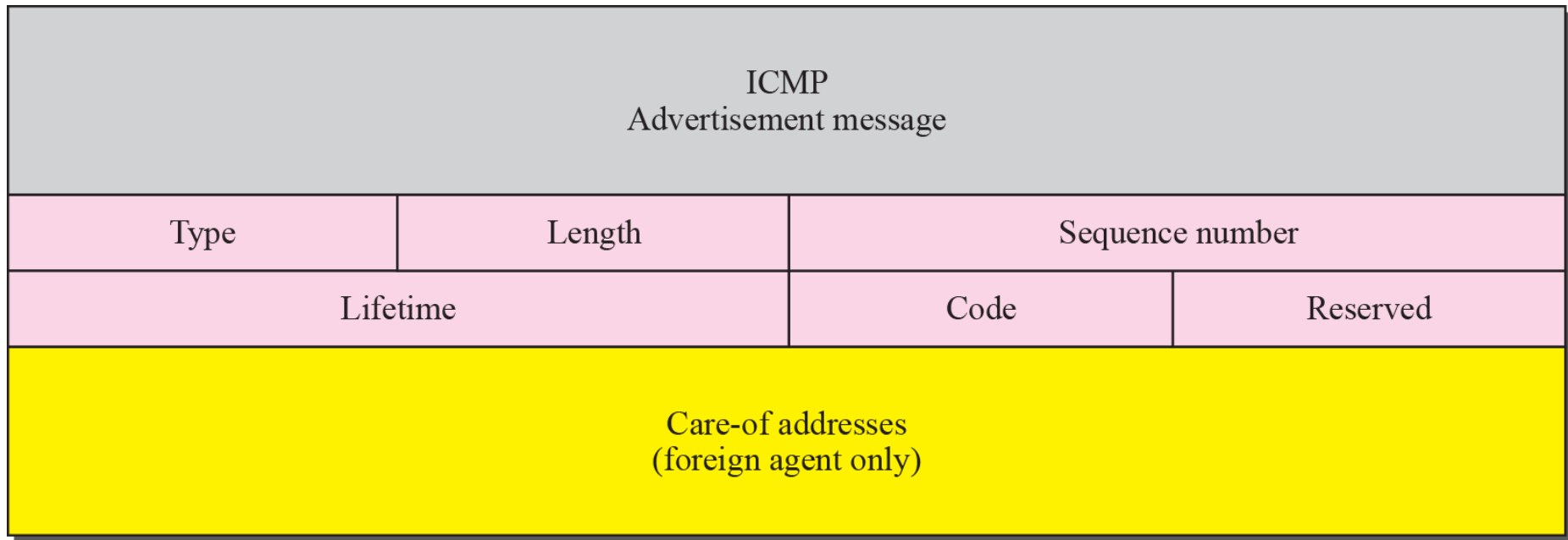
Figure 10.3 *Remote host and mobile host configuration*



Note

Mobile IP does not use a new packet type for agent advertisement; it uses the router advertisement packet of ICMP, and appends an agent advertisement message.

Figure 10.4 *Agent advertisement*



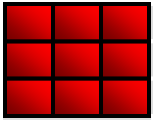


Table 10.1 *Code Bits*

<i>Bit</i>	<i>Meaning</i>
0	Registration required. No colocated care-of address.
1	Agent is busy and does not accept registration at this moment.
2	Agent acts as a home agent.
3	Agent acts as a foreign agent.
4	Agent uses minimal encapsulation.
5	Agent uses generic routing encapsulation (GRE).
6	Agent supports header compression.
7	Unused (0).

Note

Mobile IP does not use a new packet type for agent solicitation; it uses the router solicitation packet of ICMP.

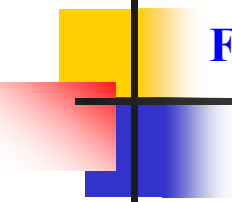


Figure 10.5 *Registration request format*

Type	Flag	Lifetime
Home address		
Home agent address		
Care-of address		
Identification		
Extensions ...		

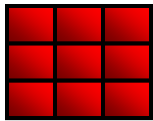


Table 10.2 *Registration request flag field bits*

<i>Bit</i>	<i>Meaning</i>
0	Mobile host requests that home agent retain its prior care-of address.
1	Mobile host requests that home agent tunnel any broadcast message.
2	Mobile host is using colocated care-of address.
3	Mobile host requests that home agent use minimal encapsulation.
4	Mobile host requests generic routing encapsulation (GRE).
5	Mobile host requests header compression.
6–7	Reserved bits.

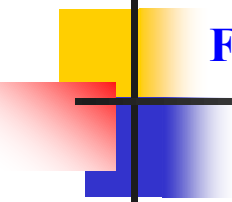


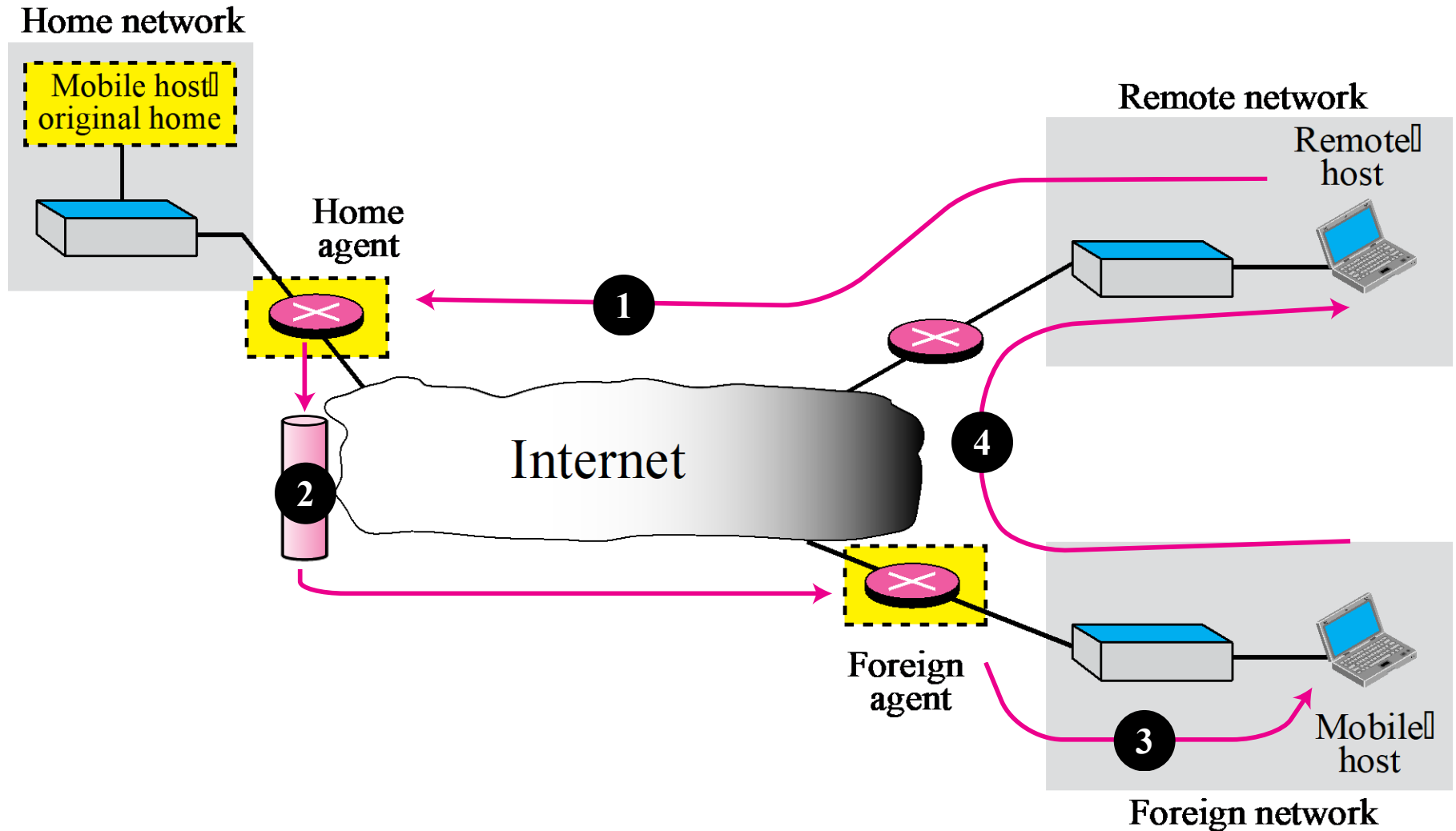
Figure 10.6 *Registration reply format*

Type	Code	Lifetime
Home address		
Home agent address		
Identification		
Extensions ...		

Note

***A registration request or reply is sent by
UDP using the well-known port 434.***

Figure 10.7 *Data transfer*



Note

The movement of the mobile host is transparent to the rest of the Internet.

10-4 INEFFICIENCY IN MOBILE IP

Communication involving mobile IP can be inefficient. The inefficiency can be severe or moderate. The severe case is called *double crossing* or *2X* . The moderate case is called *triangle routing* or *dog-leg routing*.

Topics Discussed in the Section

- ✓ **Double Crossing**
- ✓ **Triangle Routing**
- ✓ **Solution**

Figure 10.8 *Double crossing*

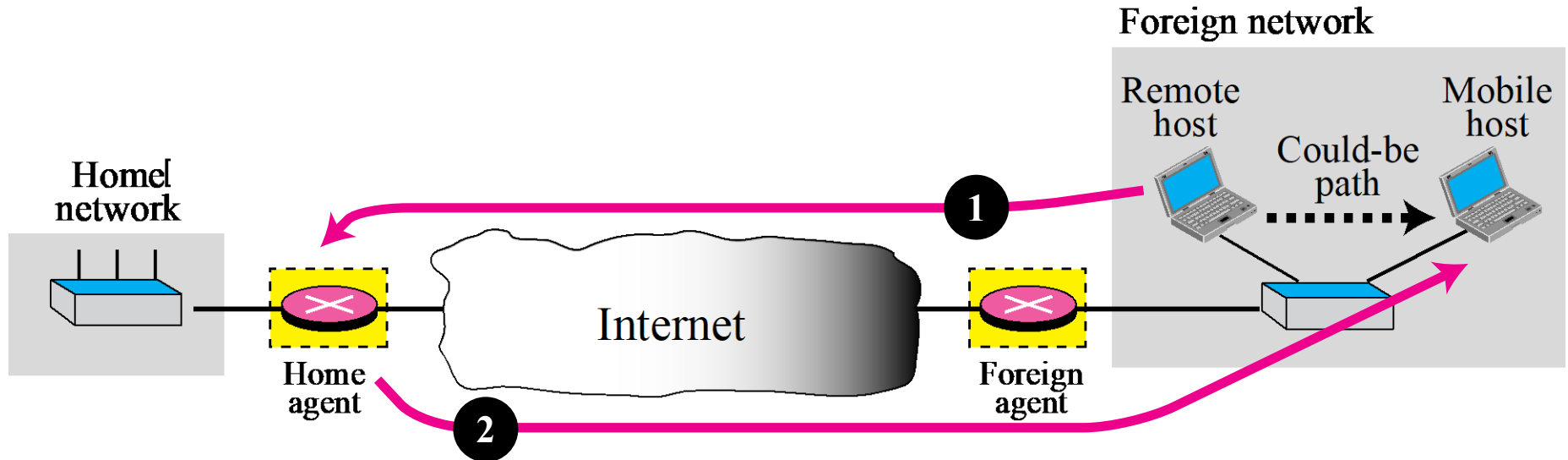


Figure 10.9 *Triangle routing*

